

LABS DOCUMENTATION

46 total

All

Standalone

Course Workspaces

Course Lecture Labs

Azure OpenAI & Core Services



Azure OpenAI and Core Services

For Azure in-curriculum (in-course) labs that need access to Azure AI services in addition to the standard Azure environment. This template is a superset of the [Azure](#) template — everything in Azure plus Azure OpenAI and Azure AI Search.

Only available in Course Lecture Labs. Modular labs and workspaces use the standard [Azure](#) template.

Policy Restrictions

Our templates map to policies that allow only the necessary services. This template adds the Microsoft.CognitiveServices and Microsoft.Search providers to the standard Azure allowlist, plus a model allowlist, a per-deployment capacity cap, and an account-kind allowlist for Azure OpenAI. Only `kind: OpenAI` Cognitive Services accounts may be created — `kind: AIServices` (multi-service Foundry) and all other kinds (Computer Vision, Speech, Document Intelligence, etc.) are blocked at account creation. Pick this template only if your lab actually needs Azure OpenAI or AI Search — the standard Azure template is preferable otherwise.

Access to VMs

When a VM is launched, Azure will set up a public IP it. However, some of our students are behind firewalls, so please use DNS names to provide access to VMs. To assign a DNS name to a VM, click on the “Configure” link to the right of DNS name as show below.



Networking

Public IP address	23.99.95.78
Public IP address (IPv6)	-
Private IP address	10.0.0.4
Private IP address (IPv6)	-
Virtual network/subnet	Regroup_4sxeOSMUvgs-vnet/default
DNS name	Configure

This will bring up the following dialog. Add a hostname as shown (but pick a hostname that better describes the lab's purpose.) The hostname needs to be unique, so also ask students to append a random number to the name. IE: springlab837102

Save
Discard
Refresh

IP address assignment

☒ Dynamic
 ☐ Static

IP address ⓘ

23.99.95.78

Idle timeout (minutes) ⓘ

4

DNS name label (optional) ⓘ

test123 ✓

.westus.cloudapp.azure.com

ⓘ You can use the IP address as your 'A' DNS record or DNS label as your 'CNAME' record. [Learn more about adding a custom domain to this IP address](#) ⓘ

Alias record sets

Create an alias record in Azure DNS. [Learn more](#) ⓘ

[+ Create alias record](#)

Subscription	DNS zone	Name	Type	TTL
No results.				

Need help?

[Using custom domains with your IP address](#) ⓘ
[Secure a web server on a Linux Virtual Machine with TLS/SSL](#) ⓘ
[Secure a web server on a Windows Virtual Machine with TLS/SSL](#) ⓘ

VM Size Recommendations

We recommend B2s as a general purpose VM size. If your lab needs more memory, a B2ms is recommended. If you need more consistent CPU performance, we suggest a D2s_v3.

Instance Type	vCPUs	RAM (GB)	Startup time (sec)	Hourly cost
B2s	2	4	35	\$

Instance Type	vCPUs	RAM (GB)	Startup time (sec)	Hourly cost
B2ms	2	8	34	\$\$
D2s_v3	2	8	45	\$\$\$\$

Supported Azure AI Services

These services are only available in this template:

- Azure OpenAI
- Azure AI Search

Supported Standard Azure Services

All of the standard Azure services are also available:

- AKS (Azure Kubernetes Service)
- API Management
- App Configuration
- App Service
- Application Gateway
- Application Insights
- Azure Functions
- Azure ML
- Azure SQL Database
- Bot Service
- CDN
- Container Instances
- Container Registry
- Cosmos DB
- Data Factory
- Event Grid
- Event Hubs
- Front Door
- Key Vault

- Load Balancer
- Log Analytics
- Logic Apps
- Managed Identity
- MySQL (Flexible Server)
- Network Security Groups
- Private Endpoints
- Public IP
- RBAC / Role Assignments
- Service Bus
- Storage (Blob, File, Queue, Table)
- Stream Analytics
- Synapse Analytics
- Virtual Machines
- Virtual Network

Service Constraints

Virtual Machines

- Allowed sizes: `Standard_A0`, `Standard_A1_v2`, `Standard_B1s`, `Standard_B1ms`, `Standard_B2s`, `Standard_B2ms`, `Standard_D1_v2`, `Standard_D2`, `Standard_D2s_v3`, `Standard_DS1`, `Standard_DS1_v2`
- No GPU sizes (ND/NC/H-series are blocked) — labs cannot self-host model serving on a GPU VM. All inference must go through Azure OpenAI deployments.
- Managed disks: `Premium_LRS`, `Standard_LRS`, `StandardSSD_LRS` only; max 1024 GB per disk

AKS (Azure Kubernetes Service)

- Node pool VM sizes: same as Virtual Machines list above, plus `Standard_DS2_v2`
- Max 3 nodes per agent pool (including `maxCount` for autoscaling)

Container Instances

- Max 3 CPU cores per container
- Max 8 GB memory per container

Cosmos DB

- Max 1,000 RU/s provisioned throughput per database/container
- Autoscale is not available

Azure Firewall

- Standard tier only (no Premium)

CDN

- Standard SKUs only: `Standard_Microsoft`, `Standard_Verizon`, `Standard_Akamai`

Event Hubs

- Namespace SKUs: `Standard`, `Basic` (no Premium / Dedicated)

Azure OpenAI — Allowed Models

Only the following base models may be deployed via Azure OpenAI

(`Microsoft.CognitiveServices/accounts/deployments`):

Base model	Type	Notes
<code>gpt-5-mini</code>	Chat	Up to 15,000 TPM per deployment
<code>gpt-5-nano</code>	Chat	Up to 15,000 TPM per deployment
<code>text-embedding-3-small</code>	Embedding	Up to 15,000 TPM per deployment

Important: All other foundation models are blocked. Labs requiring chat must use `gpt-5-mini` or `gpt-5-nano`. The embedding model is `text-embedding-3-small`. Deployment names are not constrained — you choose them freely.

Deployment SKU constraints

- Allowed SKU families: `Standard`, `GlobalStandard`, `DataZoneStandard` (TPM-based only)
- `ProvisionedManaged` and other PTU-based SKUs are blocked
- `sku.capacity` must be ≤ 15 (= 15,000 TPM per deployment)

Responsible AI

Azure RAI policies, blocklists, tool labels, and topics may be configured on Cognitive Services accounts. Instructors are responsible for applying a content filter where appropriate — none is mandated by policy.

Agent Labs

For agent-based labs, use the Azure OpenAI Assistants API on a `kind: OpenAI` account. It supports:

- Agent creation, threads, runs
- File search (vector stores)
- Code interpreter (model-dependent)
- Function calling

Azure AI Foundry's **Agent Service** (multi-agent orchestration, capability hosts, projects) is **not** available because it requires a `kind: AIServices` account, which the policy blocks. The Foundry portal at `ai.azure.com` is still usable for managing OpenAI accounts.

RAG Implementation Options

For Retrieval-Augmented Generation (RAG) pipelines, the following options are available within the sandbox:

Vector store / search backend

Option	Service	Notes
Azure AI Search	<code>Microsoft.Search/searchServices</code>	Recommended for RAG; supports vector and hybrid search
Cosmos DB for MongoDB (vCore)	<code>Microsoft.DocumentDB/databaseAccounts</code>	Native vector indexes; capped at 1,000 RU/s
Cosmos DB for NoSQL (vectorIndex)	<code>Microsoft.DocumentDB/databaseAccounts</code>	Native vector indexes; capped at 1,000 RU/s

Azure Database for PostgreSQL (pgvector) is **not** available in the sandbox.

Document storage & ingestion

Component	Service
Raw document storage	Storage (Blob)
ETL / chunking pipelines	Azure Functions, Logic Apps, Data Factory
Structured metadata queries	Azure SQL Database, Cosmos DB
Embedding generation	Azure OpenAI (<code>text-embedding-3-small</code>)
Orchestration / API layer	App Service / Azure Functions + API Management
Secrets	Key Vault
Identity / auth	Managed Identity

Reference architecture

```

1 Storage (documents)
2   → Functions / Data Factory [chunking + embedding via text-embedding-3-small]
3     → Azure AI Search [vector index]
4       → Functions [retrieval + prompt assembly]
5         → Azure OpenAI gpt-5-mini [generation]
6           → API Management [expose to client]

```

Known Problems

Azure key vault

Assigning a role (IE: admin) to the learner's account is not possible because of an autocomplete search problem on the Azure's **Add role assignment** screen. When you try to **Select member**, a sidebar dialog appears with a user search form. Even if you paste the exact username, the autocomplete search returns no results. Hence, you can not complete the role assignment.

Azure Cloud Console

When launching the Azure Cloud Console, you have several storage options:

1. **No storage account required:** Recommended. This option uses ephemeral storage for the Azure Cloud Console.

2. Mount storage account: This option provides additional choices:

- **Select existing storage account:** Not recommended. Requires an already created storage account.
- **We will create a storage account for you:** Not supported. Azure attempts to provision a storage account with a new resource group, but learners only have access to the initially provided resource group.
- **I want to create a storage account:** Not recommended. Requires additional instructions to create storage account under the resource group given at lab launch.

What is NOT Available

- Azure Database for PostgreSQL (Flexible Server or Single Server)
- Azure Database for MariaDB
- Azure Red Hat OpenShift (ARO)
- HDInsight
- GPU VM sizes (ND-, NC-, H-series)
- Azure ML serverless endpoints and Models-as-a-Service marketplace subscriptions
- PTU-based Azure OpenAI deployment SKUs (e.g., `ProvisionedManaged`)
- Any Azure OpenAI base model not listed in the allowed models table above
- Azure AI Foundry Agent Service (multi-agent orchestration, projects, capability hosts) — use the OpenAI Assistants API instead
- Azure AI Foundry multi-service accounts (`kind: AIServices`) and the Foundry Templates gallery
- Computer Vision, Document Intelligence, Speech, Translator, Content Safety, Face, Custom Vision (any Cognitive Services kind other than OpenAI)